

Q7> $x(t) = A \sin(\omega_0 t + \theta)$ $\theta \in (\pi, -\pi)$ $f_0 = 1/2\pi$

$$E[x(t)] = E[A \sin(\omega_0 t + \theta)]$$

$$= A E[\sin(\omega_0 t + \theta)]$$

$$= \frac{A}{2\pi} \int_{-\pi}^{\pi} \sin(\omega_0 t + \theta)$$

$$= \frac{A}{2\pi} [-\cos(\omega_0 t + \pi) + \cos(\omega_0 t - \pi)]$$

$$= \frac{A}{2\pi} [\cos(\omega_0 t) - \cos(\omega_0 t)]$$

$$= \frac{A}{2\pi} \cdot 0 = 0 \quad \therefore \text{Its a constant}$$

$$R_x(t, J) = E[x(t) \cdot x(t+J)]$$

$$= E[A \sin(\omega_0 t + \theta) \cdot A \sin(\omega_0 (t+J) + \theta)]$$

$$= A^2 E[\sin(\omega_0 t + \theta) \cdot \sin(\omega_0 (t+J) + \theta)]$$

$$= \frac{A^2}{2} E[\cos(\omega_0 J) - \cos(2\omega_0 t + \omega_0 J + 2\theta)]$$

$$= \frac{A^2 \cos(\omega_0 J)}{2} - \frac{A^2}{2} \int_{-\pi}^{\pi} \cos(2\omega_0 t + \omega_0 J + 2\theta)$$

$$= \frac{A^2 \cos(\omega_0 J)}{2} - \frac{A^2}{4\pi} \left(\cos(2\omega_0 t + \omega_0 J + 2\theta) - \sin(2\omega_0 t + \omega_0 J - 2\theta) \right)$$

$$= \frac{A^2 \cos(\omega_0 J)}{2} - \frac{A^2}{8\pi} (\sin(2\omega_0 t + \omega_0 J) - \sin(2\omega_0 t + \omega_0 J))$$

$$= \frac{A^2 \cos(\omega_0 J)}{2} - \frac{A^2}{8\pi} \cdot 0$$

$$= \frac{A^2 \cos(\omega_0 J)}{2}$$

$\therefore$ Its only a function of J

Hence its a wide sense stationary.

Unit - 3

Types of R.P. {WSS & SSS}

A r.p. $\{X(t)\}$ is wide sense stationary if following conditions are satisfied

- 1) $E[X(t)] = \text{a constant}$
- 2) $E[X(t) \cdot X(t+T)]$ a function of T ($t_1 - t_2$)

A RP X of t defined by $\cos(\Omega t)$ where Ω is uniform RV in $(0, \omega_0)$. check whether is stationary.

Ans: $f_0 = 1/\omega_0$

$$E[X(t)] = E[\cos(\Omega t)] \cdot 1/\omega_0$$

$$= \frac{1}{\omega_0} \int_0^{\omega_0} \cos \Omega t \, d\Omega$$

$$= \frac{1}{\omega_0} [\sin \omega_0 t - \sin 0]$$

$$= \frac{1}{\omega_0} [2 \cos \left(\frac{\omega_0 t}{2} \right) \sin \left(\frac{\omega_0 t}{2} \right)]$$

$$= \frac{\sin(\omega_0 t)}{\omega_0}$$

Not a constant but function of time t
 $\therefore X(t)$ is not stationary

A RP is defined by $\cos(t+\phi)$ where ϕ is a RV with pdf $(\phi) = 1/\pi$ from $-\pi/2$ to $\pi/2$

$$E[X(t)] = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \cos(t+\phi) \, d\phi$$

$$= \frac{1}{\pi} [\sin(t+\pi/2) - \sin(t-\pi/2)]$$

$$= \frac{1}{\pi} [2 \cos(t) \sin(\pi/2)]$$

$$= \frac{2 \cos(t) \sin(90)}{\pi} = \frac{2 \cos(t) \cdot 1}{\pi} = \frac{2 \cos(t)}{\pi}$$

not a constant but function of time

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$$\begin{aligned}
 Q10) \quad E[Y] &= E[X(2) - X(4)] \\
 E[Y^2] &= E[(X(2) - X(4))^2] \\
 &= E[(X(2))^2 + (X(4))^2 - 2(X(2)X(4))] \\
 &= 2A - 2A e^{-\kappa|2|} \\
 &= 2A(1 - e^{-2\kappa})
 \end{aligned}$$

$$\begin{aligned}
 Q11) \quad E[X(t)] &= E[X(t_1) + X_2(t_2)] \\
 &= E[X_1(t_1)] + E[X_2(t_2)] \\
 &= a + b \\
 R_x(t_1, t_2) &= E[(X_1(t_1) + X_2(t_1)) * (X_1(t_2) + X_2(t_2))] \\
 &= E[X_1(t_1)X_1(t_2) + X_1(t_1)X_2(t_2) + X_2(t_1)X_1(t_2) + X_2(t_1)X_2(t_2)] \\
 &= a^2 + a b t_1 + b b t_1 a + (a + b t_2 + t_2 b - a + 2a + 2b t_2) \\
 &= 4a^2 + -4ab t_2 + 4ab t_1 + 4b^2 b_1 t_2 \\
 C_x(t_1, t_2) &= R_x(t_1, t_2) - E[X(t_1) * X(t_2)] \\
 &= a^2 + ab t_2 + b b_1 + b^2 t_1 t_2 \\
 &= 4a^2 - 4ab t_2 + 4ab t_1 + 4b^2 t_1 t_2 - a^2 - ab t_2 - ab t_2 - b^2 t_1 t_2 \\
 &= 3(a^2 + ab t_2 + ab t_1 + b^2 t_1 t_2)
 \end{aligned}$$

$$Q12) \quad X(t) = \sin(\omega t + Y) \quad Y \in (-\pi, \pi) \quad f_0 = 1/2\pi$$

$$\begin{aligned}
 E[X(t)] &= E[\sin(\omega t + Y)] \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \sin(\omega t + Y) \\
 &= \frac{1}{2\pi} [-\cos(\omega t + 180) + \cos(\omega t - 180)] \\
 &= \frac{1}{2\pi} [\cos(\omega t) - \cos(\omega t)] \\
 &= 1/2\pi * (0) = 0 \quad \therefore \text{Its constant}
 \end{aligned}$$

$$\begin{aligned}
 R_x(t, J) &= E[\sin(\omega t + Y) \cdot \sin(\omega(t+J) + Y)] \\
 &= \frac{1}{2} E[\cos(\omega J) - \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(2\omega t + \omega J + 2Y)] \\
 &= \frac{\cos(\omega J)}{2} - \frac{1}{8\pi} [\sin(2\omega t + \omega J + 360) - \sin(2\omega t + \omega J - 360)] \\
 &= \frac{\cos(\omega J)}{2} - \frac{1}{8\pi} (\sin(2\omega t + \omega J) - \sin(2\omega t + \omega J)) \\
 &= \frac{\cos(\omega J)}{2} \quad \therefore J \text{ independent function} \\
 &\quad \therefore \text{Its wide sense stationary}
 \end{aligned}$$

STPA Assignment 10

Q1>

$$\lambda = \frac{60}{60} = 1 \quad P(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$i) P(x=2) = \frac{e^{-1} \cdot 1^2}{2!} = \frac{0.3679}{2} = 0.1839$$

$$ii) P(x=0) + P(x=1) + P(x=2)$$

$$P(x=0) = \frac{e^{-1} \cdot 1^0}{0!} = 0.3679$$

$$P(x=1) = \frac{e^{-1} \cdot 1^1}{1!} = 0.3679$$

$$: 0.3679 + 0.3679 + 0.1839 = 0.9197$$

Q2>

$$\lambda = \frac{30}{60} = 0.5 \quad t = 3$$

$$i) P(x=0) = \frac{e^{-\lambda t} \cdot (\lambda t)^x}{x!}$$

$$: \frac{e^{-(3 \times 0.5)} \cdot 1}{1} = 0.2231$$

$$ii) P(x=1) = \frac{e^{-(3 \times 0.5)} \cdot 0.5^1}{1!} = 0.2231$$

$$P(x=2) = \frac{e^{-(3 \times 0.5)} \cdot (0.5)^2}{2!} = 0.25666$$

$$P(x=3) = \frac{e^{-(3 \times 0.5)} \cdot (0.5)^3}{3!} = 0.2138$$

$$P(x=4) = 0.1336$$

$$P(x=0) = 0.2231$$

$$P(x \leq 4) = 1 - (P(0) + P(1) + P(2) + P(3) + P(4))$$

$$= 1 - (0.8912)$$

$$= 0.1088$$

STPA Assignment 8

Q1) $X(t) = \cos(\omega_0 t + \phi)$ $\phi \in (0, 2\pi)$

$f_0 = 1/2\pi$

$$E[X(t)] = E[\cos(\omega_0 t + \phi)]$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \cos(\omega_0 t + \phi) d\phi$$

$$= \frac{1}{2\pi} [\sin(\omega_0 t + 360) - \sin(\omega_0 t + 0)]$$

$$\bar{X}_T = \frac{1}{2T} \int_{-T}^T X(t) dt$$

$$= \frac{1}{2T} \int_{-T}^T \cos(\omega_0 t + \phi) dt$$

$$= \frac{1}{2T \cdot \omega_0} (\sin(\omega_0 T + \phi) - \sin(-\omega_0 T + \phi))$$

$$= \frac{1}{2T \omega_0} 2 \cos(\phi) \sin(\omega_0 T)$$

as $T \rightarrow \infty$

$$\lim_{T \rightarrow \infty} \bar{X}_T = \frac{1}{2T} \int_{-T}^T X(t) dt = 0$$

$$= 0$$

As $E[X(t)] = \bar{X}_T$ its ergodic mean

Q2) $X(t) = 10 \cos(100t + \phi)$ $\phi \in (0, 2\pi)$ $f_0 = 1/2\pi$

$$E[X(t)] = E[10 \cos(100t + \phi)]$$

$$= \frac{10}{2\pi} \int_0^{2\pi} \cos(100t + \phi) d\phi$$

$$= \frac{10}{2\pi} [\sin(100t + 360) - \sin(100t)]$$

$$= \frac{10}{2\pi} (\sin(100t) - \sin(100t))$$

$$= 0$$

classmate

Date _____
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s wandering
compartments
shown in figure.
E1 and E2 it
in 1 compartment
random. While

	E1	2	3	4	5	6	7	E2
E1	1	0	0	0	0	0	0	0
2	0.5	0	0	0.5	0	0	0	0
3	1/3	0	0	1/3	1/3	0	0	0
4	0	1/3	1/3	0	0	1/3	0	0
5	0	0	1/3	0	0	1/3	1/3	0
6	0	0	0	1/3	1/3	0	0	1/3
7	0	0	0	0	0.5	0	0	0.5
E2	0	0	0	0	0	0	0	1

Initial condition

TRM

TRM

UCV = 0.2

UC2 = 0.5

UC3 = 0.3

$$\left. \begin{array}{l} UCV = 0.2 \\ UC2 = 0.5 \\ UC3 = 0.3 \end{array} \right\} \begin{array}{l} 0.2 \\ 0.5 \\ 0.3 \end{array}$$

$$\begin{bmatrix} 0.2 & 0.2 & 0.2 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.6 & 0.4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.13 & 0.46 & 0.41 \end{bmatrix}$$

steady state probabilities / Limiting probabilities

$$P = \begin{bmatrix} 0.1 & 0.3 \\ 0.6 & 0.4 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.67 & 0.37 \\ 0.66 & 0.34 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.667 & 0.333 \\ 0.666 & 0.334 \end{bmatrix}$$

$$\underline{P} = \begin{bmatrix} 2 & 1 \\ 3 & 3 \end{bmatrix}$$

$$\bar{x}_T = \frac{1}{2T} \int_{-T}^T \theta \, dt$$

$$= \frac{1}{2T} [\theta T]_{-T}^T$$

$$= \frac{\theta}{2T} [T - (-T)]$$

$$= \frac{\theta}{2T} 2T$$

$$= \underline{\underline{\theta}}$$

$\therefore$ As $\bar{x}_T \neq E[x(t)]$ hence not ergodic

RUP

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Q8) $X(t) = \sin(\omega_0 t + \phi)$ $\phi \in (0, 2\pi) = \frac{1}{2\pi}$

$$E[X(t)] = E[\sin(\omega_0 t + \phi)]$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \sin(\omega_0 t + \phi) d\phi$$

$$= \frac{1}{2\pi} [-\cos(\omega_0 t + 360) + \cos(\omega_0 t)]$$

$$= \frac{1}{2\pi} [-\cos(\omega_0 t) + \cos(\omega_0 t)]$$

$$= \frac{1}{2\pi} [0]$$

$$= 0 \quad \therefore \text{Its a constant}$$

$$R_x(t, J) = E[X(t) \cdot X(t+J)]$$

$$= E[\sin(\omega_0 t + \phi) \cdot \sin(\omega_0(t+J) + \phi)]$$

$$= \frac{1}{2} E[\cos(\omega_0 J) - \cos(2\omega_0 t + \omega_0 J + 2\phi)]$$

$$= \frac{\cos(\omega_0 J)}{2} - \frac{1}{2} \cdot \frac{1}{2\pi} \int_0^{2\pi} \cos(2\omega_0 t + \omega_0 J + 2\phi) d\phi$$

$$= \frac{1}{2\pi} [\sin(2\omega_0 t + \omega_0 J + 2\pi) - \sin(2\omega_0 t + \omega_0 J)]$$

$$= \frac{1}{2\pi} (\sin(2\omega_0 t + \omega_0 J) - \sin(2\omega_0 t + \omega_0 J))$$

$$= \frac{1}{2\pi} (0)$$

$$= \frac{\cos(\omega_0 J)}{2}$$

Q9) $X(t) = A e^{(\omega_0 t + \phi)}$, given $E[A]$, ω_0, ϕ are constants

$$E[X(t)] = E[A e^{(\omega_0 t + \phi)}]$$

$$= e^{(\omega_0 t + \phi)} \cdot E[A]$$

$$= e^{(\omega_0 t + \phi)} \cdot 0 = 0 \quad \therefore \text{Its a constant}$$

$$R_x(t, J) = E[X(t) \cdot X(t+J)]$$

$$= E[A e^{(\omega_0 t + \phi)} \cdot A e^{(\omega_0(t+J) + \phi)}]$$

$$= e^{\omega_0 t} \cdot e^{\omega_0 t} \cdot e^{\omega_0 J} \cdot e^{\omega_0 J} \cdot E[A^2]$$

$$= e^{2\omega_0 t + 2\omega_0 J} \cdot E[A^2]$$

$\therefore$ Not only Function of J
Hence not stationary.

Q A RP is given by $x(t) = A \cos(\omega t) + B \sin(\omega t)$
 where A and B are RV such that
 $E[A] = 0$ and $E[B] = 0$, $E[A \cdot B] = 0$
 $E[A^2] = E[B^2]$
 prove its mean ergodic

$$\begin{aligned} E[x(t)] &= E[A \cos(\omega t) + B \sin(\omega t)] \\ &= \cos(\omega t) \cdot E[A] + \sin(\omega t) \cdot E[B] \\ &= \cos(\omega t) \cdot 0 + \sin(\omega t) \cdot 0 \\ &= 0 \end{aligned}$$

$$= \frac{1}{2T} \int_{-T}^T x(t) dt$$

$$= \frac{1}{2T} \int_{-T}^T A \cos(\omega t) + B \sin(\omega t) dt$$

$$= \frac{1}{2T} \left[\int_{-T}^T A \cos \omega t dt + \int_{-T}^T B \sin \omega t dt \right]$$

$$= \frac{1}{2T} \left[\frac{A}{\omega} [\sin(T) - \sin(-T)] + \frac{B}{\omega} [-\cos(T) + \cos(-T)] \right]$$

$$\frac{1}{2T} \left[\frac{A}{\omega} 2\sin(T) - \frac{B}{\omega} 2\cos(T) \right]$$

$$\frac{1}{2T\omega} [A 2\sin(T) - B 2\cos(T)]$$

as $T \rightarrow \infty$

$$\therefore \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt = 0$$

$$= 0$$

As $E[x(t)] = \text{average time}$ hence ergodic mean

3 4 5
0 0 0
0 0

1/1

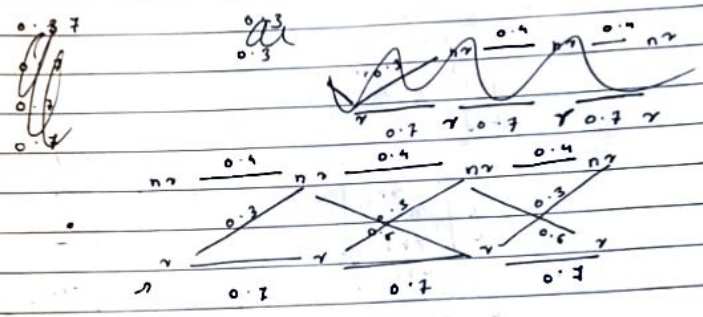
Q4)

$$P = \begin{bmatrix} a & 1-a \\ b & 1-b \end{bmatrix}$$

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.7 & 0.6 \end{bmatrix}$$

0 - rain
 1 - not rain

Q5)



$$P^k = \begin{bmatrix} 0.5749 & 0.4251 \\ 0.5668 & 0.4332 \end{bmatrix}$$

Q5)

$$P = \begin{bmatrix} 0.7 & 0.5 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0.4 \\ 0.3 & 0.4 & 0 & 0.6 \\ 0.3 & 0.4 & 0 & 0.6 \end{bmatrix}$$

RVD
1/3

$$P^k = \begin{bmatrix} 0.49 & 0.12 & 0.21 & 0.18 \\ 0.35 & 0.2 & 0.15 & 0.3 \\ 0.2 & 0.12 & 0.2 & 0.48 \\ 0.1 & 0.16 & 0.1 & 0.64 \end{bmatrix}$$

$$P(S_0 + S_1) = 0.49 + 0.12$$

$$= 0.61$$

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Q13>
$$\begin{aligned} E[X(t)] &= E[A \cos(t) + B \sin(t)] \\ &= E[A \cos(t)] + E[B \sin(t)] \\ &= \cos(t) E[A] + \sin(t) E[B] \\ &= \frac{1}{2} \cdot (-1) + \frac{1}{2} \cdot (1) \\ &= 0 \end{aligned}$$

$$\begin{aligned} R_x(t, \tau) &= E[X(t) \cdot X(t+\tau)] \\ &= E[(A \cos(t) + B \sin(t)) \cdot (A \cos(t+\tau) + B \sin(t+\tau))] \\ &= E[A^2 \cos(t) \cos(t+\tau) + AB \cos(t) \sin(t+\tau) \\ &\quad + AB \sin(t) \cos(t+\tau) + B^2 \sin(t) \sin(t+\tau)] \end{aligned}$$

$$P[X(t) = x] = \frac{e^{-\lambda t} (\lambda t)^x}{x!}$$

$$\text{mean} = \lambda t, \text{ var. } \lambda t$$

$$\text{ACR } R = (k_1 t_1 + k_2 t_2) = \lambda^* t_1 t_2 + \lambda t_1$$

$$\text{ACF } G = (k_1 t_1, k_2 t_2) = \lambda t_1$$

- Q17) A bank receives 6 bad checks per day
i) 4 bad checks on any given day
ii) 10 bad checks on any 2 consecutive days
iii) $\lambda = 6, \tau = 4, t = 1$

$$P[X(1) = 4] = \frac{e^{-6} \cdot (6 \cdot 1)^4}{4!} = 0.134$$

$$\text{ii) } P[X(2) = 10] = \frac{e^{-12} \cdot (6 \cdot 2)^{10}}{10!} = 0.105$$

Q18) min between 1 & 2

Given $\lambda = 2, t = 1$

$$\text{EXPLAN} \quad f(t) = \lambda e^{-\lambda t} = 2e^{-2t}$$

$$\begin{aligned} \text{① } P(t > 1) &= 1 - \int_0^1 2e^{-2t} dt \\ &= 2 \left[\frac{e^{-2t}}{-2} \right]_0^1 = -[e^{-2}]_0^1 = [e^0 - e^{-2}] \\ &= 1 - e^{-2} = 1 - 0.135 = 0.865 \end{aligned}$$

$$\text{② } P(1 < t < 2) = \int_1^2 2e^{-2t} dt = -[e^{-2t}]_1^2 = [e^{-2} - e^{-4}] = 0.117$$

$$\text{③ } \int_0^1 2e^{-2t} dt = -[e^{-2t}]_0^1 = [e^0 - e^{-2}] = 1 - 0.135 = 0.865$$

$$\begin{aligned}\bar{x}_T &= \frac{1}{2T} \int_{-T}^T x(t) dt \\ &= \frac{1}{2T} \int_{-T}^T \cos(100t + \phi) dt \\ &= \frac{1}{2T} \left[\frac{\sin(100t + \phi)}{100} \right]_{-T}^T \\ &= \frac{1}{20T} [2 \cos(\phi) \sin(100T)]\end{aligned}$$

$$\begin{aligned}\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt &= 0 \\ &= 0\end{aligned}$$

$\therefore$ As $\bar{x}_T = E[x(t)]$ hence ergodic mean

Q4) $x(t) = A \cos(\omega_0 t + \phi)$ $\phi \in (-\pi, \pi)$ $f_0 = 1/2\pi$

$$\begin{aligned}E[x(t)] &= E[A \cos(\omega_0 t + \phi)] \\ &= A E[\cos(\omega_0 t + \phi)] \\ &= \frac{A}{2\pi} \int_{-\pi}^{\pi} \cos(\omega_0 t + \phi) d\phi \\ &= \frac{A}{2\pi} [\sin(\omega_0 t + 180) - \sin(\omega_0 t - 180)] \\ &= \frac{A}{2\pi} 2 \cos(\omega_0 t) \sin(360) \\ &= 0\end{aligned}$$

$$\begin{aligned}\bar{x}_T &= \frac{1}{2T} \int_{-T}^T x(t) dt \\ &= \frac{1}{2T} \int_{-T}^T A \cos(\omega_0 t + \phi) dt \\ &= \frac{A}{2T} \left[\frac{\sin(\omega_0 t + \phi)}{\omega_0} \right]_{-T}^T \\ &= \frac{A}{2T\omega_0} [\sin(\omega_0 T + \phi) - \sin(-\omega_0 T + \phi)] \\ &= \frac{A}{2T\omega_0} 2 \cos(\phi) \sin(\omega_0 T)\end{aligned}$$

Ergo

 $x(t) = A \cos(\omega_0 t + \phi) \quad (0, 2\pi) \quad \phi$ show that
is mean ergodic

$$\textcircled{1} E[x(t)] = E[A \cos(\omega_0 t + \phi)]$$

$$= A E[\cos(\omega_0 t + \phi)]$$

$$= A \cdot \frac{1}{2\pi} \int_0^{2\pi} \cos(\omega_0 t + \phi)$$

$$= \frac{A}{2\pi} [\sin(\omega_0 t + 360) - \sin(\omega_0 t)]$$

$$= \frac{A}{2\pi} (\sin(\omega_0 t) - \sin(\omega_0 t))$$

$$= \frac{A}{2\pi} \cdot 0 = 0$$

Time average

$$\bar{x}_T = \frac{1}{2T} \int_{-T}^T x(t) dt = \frac{1}{2T} \int_{-T}^T A \cos(\omega_0 t + \phi) dt$$

$$= \frac{A}{2T} [\sin(\omega_0 t + \phi)]_{-T}^T$$

$$= \frac{A}{2T\omega_0} (\sin(\omega_0 T + \phi) + \sin(\omega_0 T - \phi))$$

$$= \frac{A}{2T\omega_0} 2 \cos(\omega_0 T) \sin(\phi)$$

As $T \rightarrow \infty$ it turns to be 0

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T A x(t) dt = 0$$

$$\therefore = 0$$

Since $\textcircled{1}$ and $\textcircled{2}$ are equal hence ergodic

Q14) $x(t) = A \cos(2\pi t)$

$$E[x(t)] = E[A \cos(2\pi t)]$$

$$= \cos(2\pi t) E[A]$$

∴ Its not a constant and hence not wide sense stationary

Q15) $E[x(t)] = E[x_1(t) + x_2(t)]$

$$= E[x_1(t)] + E[x_2(t)]$$

$$= 1 + 1.5 = 2.5$$

$$R_x(t_1, t_2) = E[(x_1(t_1) - E x_1(t_1)) + (x_2(t_2) - E x_2(t_2))]$$

$$= E[x_1^2(t_1)(t_2) + x_1 x_2(t_1)(t_2) + x_2 x_1(t_1)(t_2) + x_2^2(t_1)(t_2)]$$

$$= E[x_1^2] + E[x_2^2] + 2E[x_1 \cdot x_2]$$

$$= E[x_1^2] + E[x_2^2] + 2(E[x_1] \cdot E[x_2])$$

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Ans
20/02

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A RP $X(t) = A \sin(\omega_0 t)$ where ω_0 is constant and A is a RV uniformly distributed over -1 to 1 prove that process is not stationary.

$$f(A) = \frac{1}{2} \quad -1 < A < 1$$

$$E[X(t)] = E[A \sin(\omega_0 t)] \cdot f_0$$

$$= \frac{1}{2} \sin(\omega_0 t) \int_{-1}^1 A \, dA$$

$$= \left[\frac{A^2}{2} \right]_{-1}^1 \cdot \frac{\sin(\omega_0 t)}{2}$$

$$= \left[\frac{1}{2} - \frac{1}{2} \right] \cdot \frac{\sin(\omega_0 t)}{2}$$

$$= 0, \text{ a constant}$$

$$E[X(t) \cdot X(t+J)]$$

$$= E[A \sin(\omega_0 t) \cdot A \sin(\omega_0 (t+J))]$$

$$= \sin(\omega_0 t) \cdot \sin(\omega_0 (t+J)) [A^2]$$

$$= \frac{\sin(\omega_0 t) \cdot \sin(\omega_0 (t+J))}{2} \cdot \left[\frac{A^3}{3} \right]_{-1}^1$$

$$= \frac{\sin(\omega_0 t) \cdot \sin(\omega_0 (t+J))}{2} \cdot \left(\frac{1}{3} - \left(-\frac{1}{3} \right) \right)$$

$$= \frac{\sin(\omega_0 t) \cdot \sin(\omega_0 (t+J))}{2} \cdot \frac{2}{3}$$

$$= (\sin(\omega_0 t) \cdot \sin(\omega_0 (t+J))) / 3$$

$\Rightarrow$ not a function of J and $\therefore X(t)$ is not stationary

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show that the AP given by $x(t)$ given by $A \cos(\omega_0 t + \phi)$ where A and ω_0 are constants and ϕ is uniformly distributed over $(0, 2\pi)$ is a wide sense stationary

$$\begin{aligned}
 E[x(t)] &= E[A \cos(\omega_0 t + \phi)] \\
 &= \frac{A}{2\pi} \int_0^{2\pi} \cos(\omega_0 t + \phi) d\phi = \frac{A}{2\pi} [\sin(\omega_0 t + \phi)]_0^{2\pi} \\
 &= \frac{A}{2\pi} [\sin(\omega_0 t + 360) - \sin(\omega_0 t - 360)] \\
 &= \frac{A}{2\pi} [2 \cos(2\omega_0 t) \sin(720)] \\
 &= \frac{A}{2\pi} \cos(\omega_0 t) \cdot 0 = 0, \text{ a constant}
 \end{aligned}$$

$$\begin{aligned}
 R_x(J) &= E[A \cos(\omega_0 t + \phi) \cdot A \cos(\omega_0(t+J) + \phi)] \\
 &= A^2 E[\cos(\omega_0 t + \phi) \cos(\omega_0(t+J) + \phi)]
 \end{aligned}$$

$$= A^2 \left[\frac{1}{2} [\cos(\omega_0 t + \omega_0 t + \omega_0 J + 2\phi) + \cos(\omega_0 J)] \right]$$

$$= \frac{A^2}{2} \cos(\omega_0 J) + \frac{1}{2} A^2 \int_0^{2\pi} \cos(2\omega_0 t + \omega_0 J + 2\phi) \cdot \frac{1}{2\pi} d\phi$$

$$= \frac{A^2}{2} \cos(\omega_0 J) + \frac{A^2}{4\pi} \int_0^{2\pi} \cos(2\omega_0 t + \omega_0 J + 2\phi) d\phi$$

$$= \frac{A^2}{8\pi} [\sin(2\omega_0 t + \omega_0 J + 2\phi)]_0^{2\pi}$$

$$= \frac{A^2}{8\pi} [\sin(2\omega_0 t + \omega_0 J + 720) - \sin(2\omega_0 t + \omega_0 J)]$$

$$= \frac{A^2}{8\pi} 2 \cos(2\omega_0 t + \omega_0 J) \sin(360)$$

$$= \frac{A^2}{8\pi} [2 \sin(2\omega_0 t + \omega_0 J) - \sin(2\omega_0 t + \omega_0 J)]$$

$$= \frac{A^2}{8\pi} \cdot 0$$

$$= \frac{A^2}{2} \cos(\omega_0 J) = \frac{A^2}{2} \cos(\omega_0 J)$$

Evolution Process

A Process which is not stationary in any sense is called evolutionary process. In other words, a RP is evolutionary if the mean & variance are function of t hence dependent on t .

Ergodicity:

$$\bar{X}_t = \frac{1}{2T} \int_{-T}^T X(t) dt \quad \left\{ \begin{array}{l} \text{Time average} \end{array} \right.$$

when time average = ensemble average $\{ = [E[X(t)]]$

STPA Assignment 7

$$\begin{aligned} Q1) \quad E[X(t)] &= E[\cos(\omega t)] \quad f_0 = 1/1 + 0 = 1 \\ &= \int_0^1 \cos(\omega t) \cdot 1 \\ &\quad \cdot [\sin(\omega t)]_0^1 \\ &= \sin(t) - \sin(0) \\ &= \sin(t) \end{aligned}$$

∴ Not a constant but function of time t
 ∴ $X(t)$ is not stationary.

$$\begin{aligned} Q2) \quad E[X(t)] &= E[A \cos(\omega_0 t)] \quad f_0 = 1/\pi + 0 = 1/\pi \\ &= A \cdot \frac{1}{\pi} \cdot \int_0^\pi \cos(\omega_0 t + \phi) \\ &= \frac{A}{\pi} [\sin(\omega_0 t + \phi) - \sin(\phi)] \\ &= \frac{A - 2\sin(\omega_0 t)}{\pi} \end{aligned}$$

∴ Not a constant but function of time t
 ∴ $X(t)$ is not stationary

$$\begin{aligned} Q3) \quad E[X(t)] &= E[A \cos(\omega_0 t + \phi)] \quad f_0 = 1/\pi + 0 = 2/\pi \\ &= \frac{2A}{\pi} \int_0^\pi \cos(\omega_0 t + \phi) \\ &= \frac{2A}{\pi} [\sin(\omega_0 t + \pi/2) - \sin(\omega_0 t + \phi)] \\ &= \frac{2A}{\pi} (\cos(\omega_0 t) - \sin(\omega_0 t)) \end{aligned}$$

∴ Not a constant but function of time t
 ∴ $X(t)$ is not stationary

//_

$$\begin{aligned}
 Q4) E[X(t)] &= E[A \cos(\omega_0 t)] \quad f_0 = 1/10 = 1 \\
 &= \cos(\omega_0 t) \int_0^A A \\
 &= \cos(\omega_0 t) [A^2/2] \\
 &= \frac{\cos(\omega_0 t)}{2}
 \end{aligned}$$

Not a constant but function of time t
 $\therefore X(t)$ is not stationary

$$\begin{aligned}
 Q5) E[X(t)] &= E[10 \cos(100t + \phi)] \quad f_0 = 1/\pi - (-\pi) = 1/2\pi \\
 &= \frac{10}{2\pi} \int_{-\pi}^{\pi} \cos(100t + \phi) d\phi \\
 &= \frac{5}{\pi} [\sin(100t + 180) - \sin(100t - 180)] \\
 &= \frac{5}{\pi} \left(2 \cos\left(\frac{200t}{2}\right) \sin\left(\frac{360}{2}\right) \right) \\
 &= \frac{10}{\pi} \cos(100t) \sin(180) \\
 &= \frac{10}{\pi} \cos(100t) \cdot 0 = 0
 \end{aligned}$$

$\therefore$ its a constant

$$\begin{aligned}
 E[X(t) \cdot X(t+J)] &= E[10 \cos(100t + \phi) \cdot 10 \cos(100(t+J) + \phi)] \\
 &= 100 E[\cos(100t + \phi) \cdot \cos(100(t+J) + \phi)] \\
 &= \frac{100}{2} E[\cos(200t + 100J + 2\phi) + \cos(100J)] \\
 &= 50 \cos(100J) + 50 \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(200t + 100J + 2\phi) \\
 &\quad + \frac{6.25}{2\pi} [\sin(200t + 100J + 360) \cdot \sin(200t + 100J) - 360] \\
 &\quad + \frac{25}{\pi} \left[\cos\left(\frac{400t + 200J}{2}\right) \sin\left(\frac{120}{2}\right) \right] \\
 &\quad + \frac{25}{\pi} \cos(200t + 100J) \sin(360) \\
 &\quad + 0
 \end{aligned}$$

50 cos(100T)
Its a function of T
Its a WSS

RV
13/12
 $f_0 = \frac{1}{2\pi} \cdot 0 = \frac{1}{2\pi}$

Q6) $E[X(t)] = E[\sin(\omega_0 t + Y)]$
 $= \frac{1}{2\pi} \int_0^{2\pi} \sin(\omega_0 t + Y) dY$
 $= \frac{1}{2\pi} [-\cos(\omega_0 t + Y)]_0^{2\pi}$
 $= -\frac{1}{2\pi} [\cos(\omega_0 t + 360) - \cos(\omega_0 t)]$
 $= -\frac{1}{2\pi} [\cos(\omega_0 t) - \cos(\omega_0 t)]$
 $= -\frac{1}{2\pi} \cdot 0$
 $= 0$ $\therefore$ Its a constant

$R_x(t, T) = E[X(t) \cdot X(t+T)]$
 $= E[\sin(\omega_0 t + Y) \cdot \sin(\omega_0(t+T) + Y)]$
 $= \frac{1}{2} E[\cos(\omega_0 T) - \cos(2\omega_0 t + \omega_0 T + 2Y)]$
 $= \frac{\cos(\omega_0 T)}{2} - \frac{1}{2} \int_0^{2\pi} \cos(2\omega_0 t + \omega_0 T + 2Y) dY$
 $= \frac{\cos(\omega_0 T)}{2} - \frac{1}{2 \cdot 2} [\sin(2\omega_0 t + \omega_0 T + 2Y)]_0^{2\pi}$
 $= \frac{\cos(\omega_0 T)}{2} - \frac{1}{4\pi} (\sin(2\omega_0 t + \omega_0 T) - \sin(2\omega_0 t + \omega_0 T))$
 $= \frac{\cos(\omega_0 T)}{2} - 0$
 $= \frac{\cos(\omega_0 T)}{2}$

$\therefore$ Its a function of T

Hence its a WSS

Q3) $\lambda = 2$

$$P(x=0) = \frac{e^{-(2 \times 5)} \cdot (2 \times 5)^0}{0!} = e^{-10} = 4.54 \times 10^{-5}$$

Q4)

$$i) P(x=4) = \frac{e^{-(3 \times 2)} \cdot (3 \times 2)^4}{4!} = 0.133$$

$$ii) P(x=0) = \frac{e^{-(3 \times 2)} \cdot 1}{1} = 0.00248$$

$$P(x=1) = \frac{e^{-(3 \times 2)} \cdot (3 \times 2)^1}{1!} = 0.0149$$

$$P(x=2) = \frac{e^{-(3 \times 2)} \cdot (3 \times 2)^2}{2!} = 0.0446$$

$$P(x=3) = \frac{e^{-(3 \times 2)} \cdot (3 \times 2)^3}{3!} = 0.0892$$

$$P(x > 4) = 1 - (0.00248 + 0.0149 + 0.0446 + 0.0892) \\ = 1 - 0.1512 \\ = 0.8488$$

Q5) Given time interval formula is $e^{-\lambda t}$

$$i) P(T > 1) = \frac{e^{-(3 \cdot 1)}}{1} = e^{-3} = 0.0498$$

$$ii) P(1 < T < 2) = P(T > 1) - P(T > 2) \\ = e^{-3 \cdot 1} - e^{-3 \cdot 2} \\ = 0.0498 - 0.00248 = 0.0473$$

$$iii) P(T \leq 3) = 1 - P(T > 3) \\ = 1 - e^{-3 \cdot 3} \\ = 1 - e^{-9} \\ = 0.999$$

If TRM of markov chain is $P = \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix}$ find its steady state matrix upto power 8

$$P = \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.25 & 0.75 \\ 0.375 & 0.625 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} 0.375 & 0.625 \\ 0.3125 & 0.6875 \end{bmatrix}$$

$$P^5 = \begin{bmatrix} 0.3125 & 0.6875 \\ 0.3437 & 0.6562 \end{bmatrix}$$

$$P^6 = \begin{bmatrix} 0.3437 & 0.6562 \\ 0.3281 & 0.6718 \end{bmatrix}$$

$$P^7 = \begin{bmatrix}$$

$$P^8 = \begin{bmatrix} 0.3359 & 0.6641 \\ 0.332 & 0.6679 \end{bmatrix}$$

$$\pi = \left[\frac{1}{3}, \frac{2}{3} \right]$$

[Signature]

Q6)

$$P(x=10) = \frac{e^{-\lambda pt} \cdot (\lambda pt)^x}{x!}$$

$$= \frac{e^{-(6 \cdot 0.5 \cdot 4)} \cdot (6 \cdot 0.5 \cdot 4)^{10}}{10!} = 0.104$$

Q7)

i) $P(x=0) = e^{-(6 \cdot 1)} \cdot 1 = e^{-6} = 0.00248$

ii) $P(x=6) = \frac{e^{-(6 \cdot 1)} \cdot (6 \cdot 1)^6}{6!} = 0.1607$

iii) $P(x \leq 3) = P(0) + P(1) + P(2)$

$$P(x=0) = e^{-(6 \cdot 0.5)} \cdot (6 \cdot 0.5)^0 = e^{-3} = 0.0498$$

$$P(x=1) = \frac{e^{-(6 \cdot 0.5)} \cdot (6 \cdot 0.5)^1}{1!} = 3e^{-3} = 0.1494$$

$$P(x=2) = \frac{e^{-(6 \cdot 0.5)} \cdot (6 \cdot 0.5)^2}{2!} = 1.5e^{-3} = 0.2240$$

$$= 0.0498 + 0.1494 + 0.2240 = 0.4231$$

Q8)

$$\lambda = \frac{1}{3} = 0.3$$

$$P(x \leq 2) = P(0) + P(1) + P(2)$$

$$= P(x=0) = \frac{e^{-(0.2 \cdot \frac{1}{3})} \cdot 1}{0!} = e^{-0.2} = 0.8187$$

$$P(x=1) = \frac{e^{-(0.2 \cdot \frac{1}{3})} \cdot (0.2 \cdot \frac{1}{3})^1}{1!} = 0.2 \cdot e^{-0.2} = 0.1637$$

$$P(x=2) = \frac{e^{-(0.2 \cdot \frac{1}{3})} \cdot (0.2 \cdot \frac{1}{3})^2}{2!} = 0.0205$$

$$= 0.8187 + 0.1637 + 0.0205 = 1.0029$$

$$\begin{aligned}
 \bar{x}_T &= \frac{1}{2T} \int_{-T}^T x(t) dt \\
 &= \frac{10}{2T} \int_{-T}^T \cos(100t + \phi) dt \\
 &= \frac{10}{2T} \cdot \frac{1}{100} \left[\sin(100T + \phi) - \sin(-100T + \phi) \right] \\
 &= \frac{1}{20T} \left(2 \cos\left(\frac{2\phi}{2}\right) \sin\left(\frac{200T}{2}\right) \right) \\
 &= \frac{1}{20T} \cos(\phi) \sin(100T)
 \end{aligned}$$

as $T \rightarrow \infty$

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt = 0$$

As $\bar{y}_T = E[x(t)]$ hence ergodic mean.

Q3)
Q4)

$$\begin{aligned}
 X &= 10 \cos(100t + \phi) \quad \phi \in (-\pi, \pi) \quad f_0 = \frac{1}{2\pi} \\
 E[X(t)] &= E[10 \cos(100t + \phi)] \\
 &= \frac{10}{2\pi} \int_{-\pi}^{\pi} \cos(100t + \phi) d\phi \\
 &= \frac{10}{2\pi} [\sin(100t + 180) - \sin(100t - 180)] \\
 &= \frac{10}{2\pi} \cdot 2 \cos(100t) \sin(360) \\
 &= 0
 \end{aligned}$$

classmate

Date

Page

$$\begin{aligned}
 \phi &= \frac{1}{2T} \int_{-T}^T \cos(t + \phi) dt \\
 &= \frac{1}{2T} \int_{-T}^T \sin(\phi) dt \\
 &= \frac{1}{2T} [\sin(t + \phi) + \sin(-t + \phi)] \\
 &= \frac{1}{2T} (\sin(t + \phi) + \sin(-t + \phi)) \\
 &= \frac{1}{2T} 2 \cos(\phi) \sin(t) \\
 &= \frac{\cos(\phi) \sin(t)}{T}
 \end{aligned}$$

as T tends to infinity $\sin(t) = 0$

$\therefore$ Time average $\bar{x}_T = 0$

$\therefore E[x(t)] = \bar{x}_T \therefore$ Its ergodic mean

(6.0.5.4)
10.1

KRI TPM

state num

	0	1	2	3	4
0	1	0	0	0	0
1					

1/1

Q2) 1) 0.3×0.3
 $\times 0.6 \times 0.6$
 $= 0.45 \times 0.03$

2) $0.4 \times 0.5 \times 0.1$
 $0.5 \times 0.4 \times 0.3$
 $(0.2 + 0.2 + 0.03) = 0.43$
 $= 0.43$

Q3)
$$P = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 3/4 & 1/4 & 0 \\ 1 & 1/4 & 2/4 & 1/4 \\ 2 & 0 & 3/4 & 1/4 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 5/8 & 5/16 & 1/16 \\ 3/16 & 1/2 & 3/16 \\ 3/16 & 9/16 & 1/4 \end{bmatrix}$$

$P(X_2=2) = (1/3) \times (1/16) + (1/3) \times (3/16) + (1/3) \times (1/4)$
 $= \frac{6}{48} = \frac{1}{6}$

$P(X_3=1, X_2=2, X_1=1, X_0=2)$

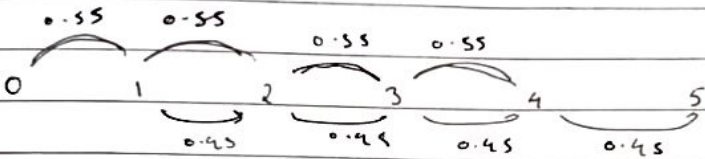
$(P)^3 = \begin{bmatrix} 35/64 & 23/64 & 6/64 \\ 23/64 & 30/64 & 11/64 \\ 18/64 & 33/64 & 13/64 \end{bmatrix}$
 $= \frac{19}{48}, \frac{1}{6}, \frac{1}{3}, \left(\frac{1}{3} \times 0 + \frac{1}{3} \times \frac{1}{4} + \frac{1}{3} \times \frac{1}{4} \right)$
 $= \left[\frac{19}{48}, \frac{1}{6}, \frac{1}{3}, \frac{2}{12} \right]$

$= P(X_3=1, X_2=2, X_1=1, X_0=2)$

$= \frac{3}{4} \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{3} = \frac{3}{64}$

A gambler wins ₹1 with 0.45 and loses ₹1 with 0.55 at any state. He starts game with ₹1 and quits it when either he has earned ₹4 or he is bankrupt.

Write transition probability matrix.



~~TPM~~ ~~TPM~~ TPM

		state $n+1$					
		0	1	2	3	4	5
state n	0	1	0	0	0	0	0
	1	0.55	0	0.45	0	0	0
	2	0	0.55	0	0.45	0	0
	3	0	0	0.55	0	0.45	0
	4	0	0	0	0.55	0	0.45
	5	0	0	0	0	0	1

Q9 > $\lambda = \frac{1}{5} \cdot 0.2$

$P(X \geq 2) = 1 - P(X \leq 2)$ $P(X \leq 2) = P(0) + P(1) + P(2)$

$P(X=0) = e^{-(0.2 \times 7)} \cdot 1 = e^{-1.4} = 0.2466$

$P(X=1) = e^{-(0.2 \times 7)} \cdot (0.2 \times 7) = 1.4 e^{-1.4} = 0.3453$

$P(X=2) = \frac{e^{-(0.2 \times 7)} \cdot (0.2 \times 7)^2}{2!} = 0.2417$

$= 1 - (0.3453 + 0.2417 + 0.2466)$
 $= 0.1665$

Q9 > $\lambda = 1$

i) $P(t > 2) = \int_2^{\infty} \lambda e^{-t}$
 $= \int_2^{\infty} e^{-t}$
 $= [-e^{-t}]_2^{\infty}$
 $= [-0 + e^{-2}] \cdot e^{-2} = 0.1353$

ii) $P(X(10) \leq 5)$

$= e^{-10} \cdot 1 + 10e^{-10} + \frac{100e^{-10}}{2!} + \frac{1000e^{-10}}{3!} + \frac{10000e^{-10}}{4!} + \frac{100000e^{-10}}{5!}$
 $= 0.067$

Q10 > $\lambda = 2$

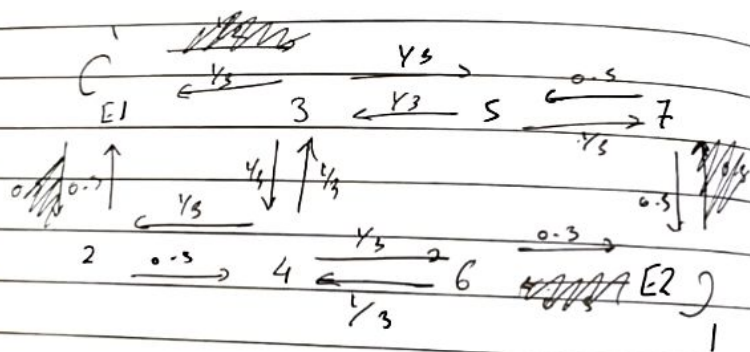
i) $P(X=1) = \frac{e^{-(2 \cdot 0.5)} \cdot (2 \cdot 0.5)^1}{1!} = 0.5 \cdot 1 \cdot e^{-1} = 0.3679$

ii) $P(X=3) = \frac{e^{-(2 \cdot 0.5)} \cdot (2 \cdot 0.5)^3}{3!} = \frac{e^{-1} \cdot 1}{6} = 0.1953$

RVP
 20/02

Q.1) Consider an exp in which a rat is wandering inside a maze which has 8 compartments. Each compartment has 2 doors as shown in figure. If rat reaches compartment E1 and E2 it escapes otherwise it moves from 1 compartment to other through the doors at random. Write transition matrix and diagram

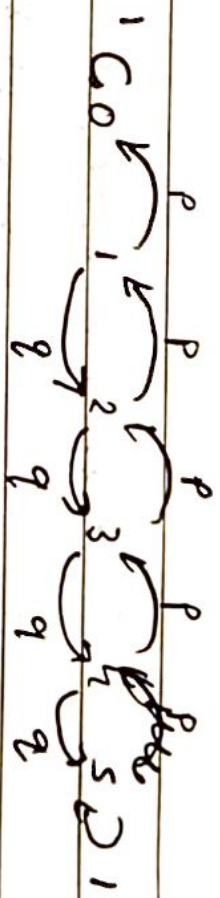
E1	3	1/3	5	1/3	7
1			1/3		1/2
2	4	1	6		E2



E1 — 2 — 3 — 4 — 5 — 6 — 7 — E2

Q2

	1	2	3	4	5
1	0	1	0	0	0
2	0	0	0	0	0
3	0	0	0	0	0
4	0	0	0	0	0
5	0	0	0	0	0



A RP given by $x(t) = \cos(\omega t + \phi)$ where ϕ is a RV ud over $(0, 2\pi)$

- ① Show that process is stationary in wide sense
 ② Show that process is mean ergodic

$$\begin{aligned} \textcircled{1} \quad & E[x(t)] \\ &= E[\cos(t + \phi)] \\ &= \frac{1}{2\pi} \int_0^{2\pi} \cos(t + \phi) \\ &= \frac{1}{2\pi} [\sin(t + 360) - \sin(t)] \\ &= \frac{1}{2\pi} (\sin(t) - \sin(t)) \\ &= \underline{0} \end{aligned}$$

$$\begin{aligned} & E[x(t) \cdot x(t+J)] = E[\cos(t + \phi) \cdot \cos(t+J + \phi)] \\ &= \frac{1}{2} E[\cos(2t + 2\phi + J) + \cos(J)] \\ &= \frac{1}{2} \cos(J) + \frac{1}{2} E \left[\frac{1}{2\pi} \int_0^{2\pi} \cos(2t + J + 2\phi) d\phi \right] \\ &= \frac{1}{2} \cos(J) + \frac{1}{2} \int_0^{2\pi} \frac{\sin(2t + J + 2\phi)}{2\pi} \cdot \sin(2\phi) \\ &= \frac{\cos(J)}{2} + 0 \end{aligned}$$

$\therefore$ cos is function of J hence WSS

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t) dt = 0$$

$$= 0$$

As $\bar{X}_T = E[X(t)] \therefore$ Its ergodic mean.

Q5) $X(t) = \theta \quad \theta \in (-1, 1) \quad \text{to } \frac{1}{2}$

$$E[X(t)] = E[\theta]$$

$$= \frac{1}{2} \int_{-1}^1 \theta$$

$$= \frac{1}{2} \left[\frac{\theta^2}{2} \right]_{-1}^1$$

$$= \frac{1}{2} \left[\frac{1}{2} - \left(-\frac{1}{2} \right) \right]$$

$$= 0$$

$$R_x(t, J) = E[X(t) \cdot X(t+J)]$$

$$= E[\theta \cdot \theta + J] = E[\theta \cdot \theta + J]$$

$$= \theta^2 E[1] = E[\theta^2 + \theta J]$$

$$= E[\theta^2] + E[\theta J]$$

$$\frac{1}{2} \left[\frac{\theta^3}{3} \right]_{-1}^1 + \frac{J}{2} \left[\frac{\theta^2}{2} \right]_{-1}^1$$

$$= \frac{1}{2} \left[\frac{1}{3} - \left(-\frac{1}{3} \right) \right] + \frac{J}{2} \left[\frac{1}{2} - \frac{1}{2} \right]$$

$$= \frac{2}{3} \cdot \frac{1}{2} + 0$$

$$= \underline{\underline{1/3}}$$

A radio active substance emits particles at the rate of 6 particles per minute according to poisson process. Prob that emitted particles will be recorded is 0.5 find prob that 10 particles will be recorded in 4 mins time period

$$\begin{aligned} P(N(t) = r) &= \frac{e^{-\lambda t} \cdot (\lambda t)^r}{r!} \\ &= \frac{e^{-6 \cdot 0.5 \cdot 4} \cdot (6 \cdot 0.5 \cdot 4)^{10}}{10!} \\ &= \underline{0.104} \end{aligned}$$